Technology Stack

Date	27 June 2026
Team ID	
Project Name	**Credit Card Approval Prediction**
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	**Frontend / Client-Side**	HTML5, CSS3, Bootstrap 5, JavaScript	Provides a responsive, user-friendly interface for entering applicant details and displaying prediction results across desktop and mobile devices.

2	**Backend / Server-Side**	Python 3.x, Flask	Flask handles HTTP requests, processes user inputs, communicates with the machine learning model, and returns prediction results efficiently
3	**Database / Data Storage**	CSV Dataset, Pickle (.pkl) Model File	CSV stores training data, while the Pickle file stores the trained machine learning model for fast prediction without retraining.
4	**Cloud / Hosting / Deployment**	IBM Cloud, IBM Code Engine (or IBM Cloud Foundry)	Provides reliable cloud deployment, scalability, easy application hosting, and online accessibility.

5	**Version Control & CI/CD**	Git, GitHub	Enables version control, collaboration, source code management, and project backup. GitHub stores the project repository securely.
6	**Third-Party APIs / Other Tools**	Scikit-learn, Pandas, NumPy, Matplotlib, Joblib	Scikit-learn builds the ML model, Pandas handles data preprocessing, NumPy performs numerical computations, Matplotlib generates visualizations, and Joblib/Pickle saves and loads the trained model efficiently.